

Chapter Six

Power Supplies, Breakdown Diodes and Voltage Regulators

6 hours, 8 marks

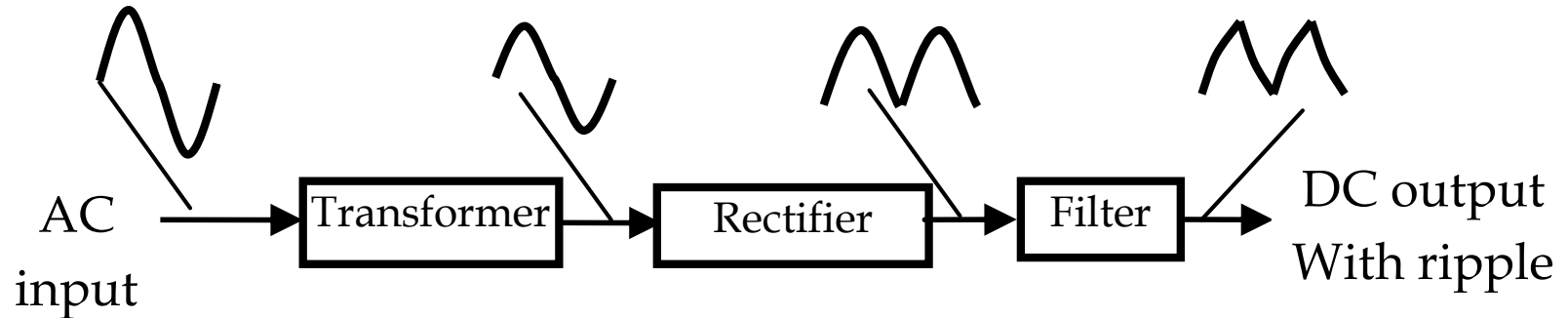
- 6.1. Unregulated Power Supply
- 6.2. Bandgap Voltage Reference, a Constant Current Diodes
- 6.3. Transistor Series Regulators
- 6.4. Improving Regulator Performance
- 6.5. Current Limiting
- 6.6. Integrated Circuit Voltage Regulator

POWER SUPPLY

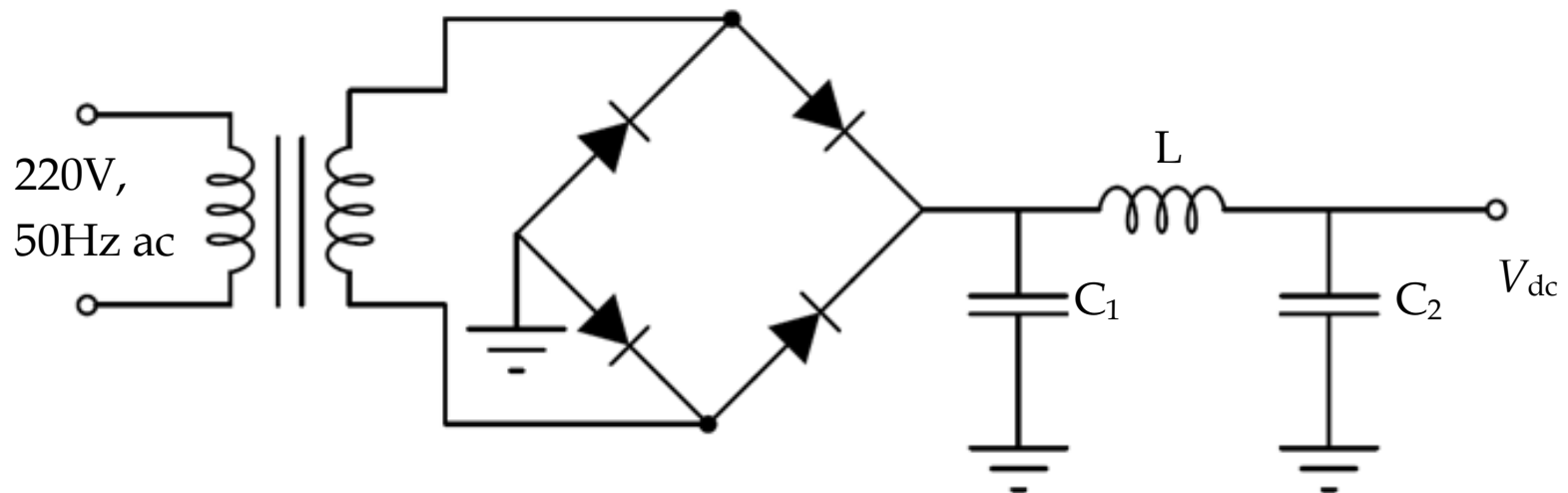
- ✓ In many applications of electronics, it is required to use DC power instead of AC power.
- ✓ The process of conversion of AC current/voltage into DC current/voltage is known as rectification.
- ✓ It is achieved through the use of rectifier system.
- ✓ **Power supply** is a complete system consisting of transformer (to step down ac voltage), rectifier (to convert ac into dc), filter (to smoothen pulsating dc) and finally a voltage regulator (to get stable dc output).

UNREGULATED DC POWER SUPPLY SYSTEM

- ✓ DC Power supply without the voltage regulator circuit at its output is known as unregulated dc power supply.
- ✓ The block diagram of unregulated DC power supply is:-

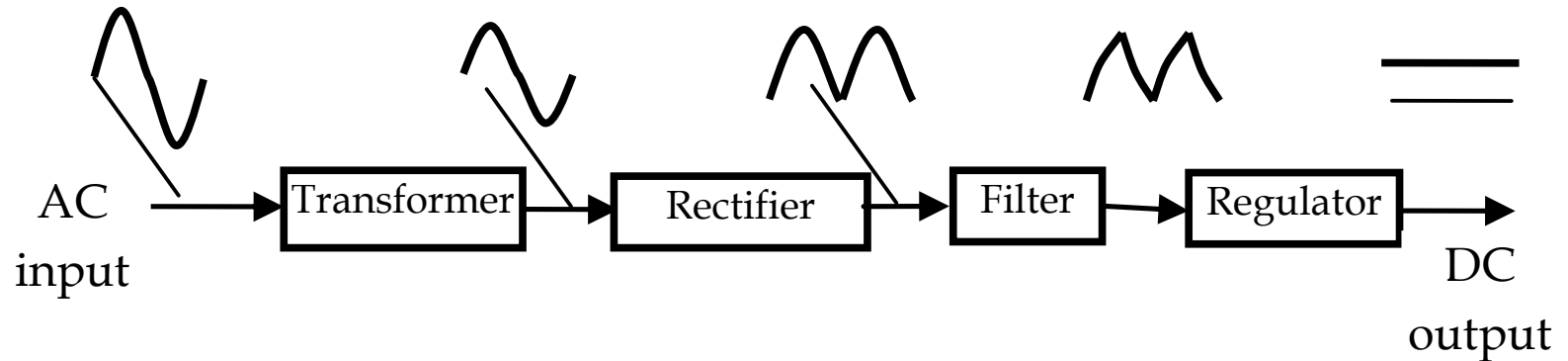


- ✓ The circuit diagram implementation of regulated DC power supply system is:

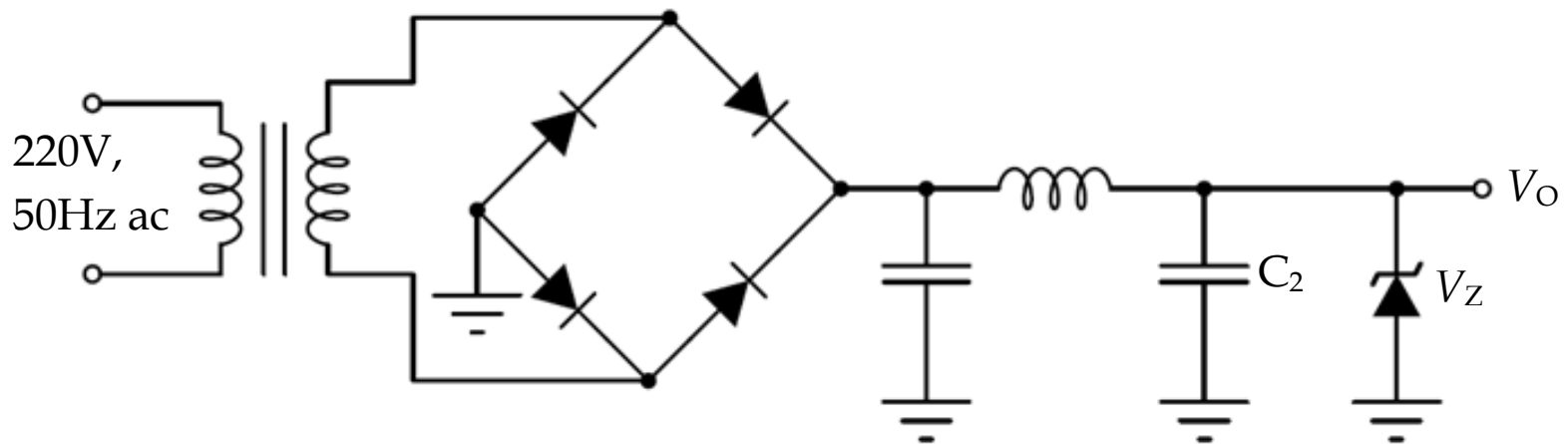


REGULATED DC POWER SUPPLY SYSTEM

- ✓ The DC power supply system with the voltage regulator at the output is known as regulated dc power supply.
- ✓ The block diagram of regulated DC power supply system is:

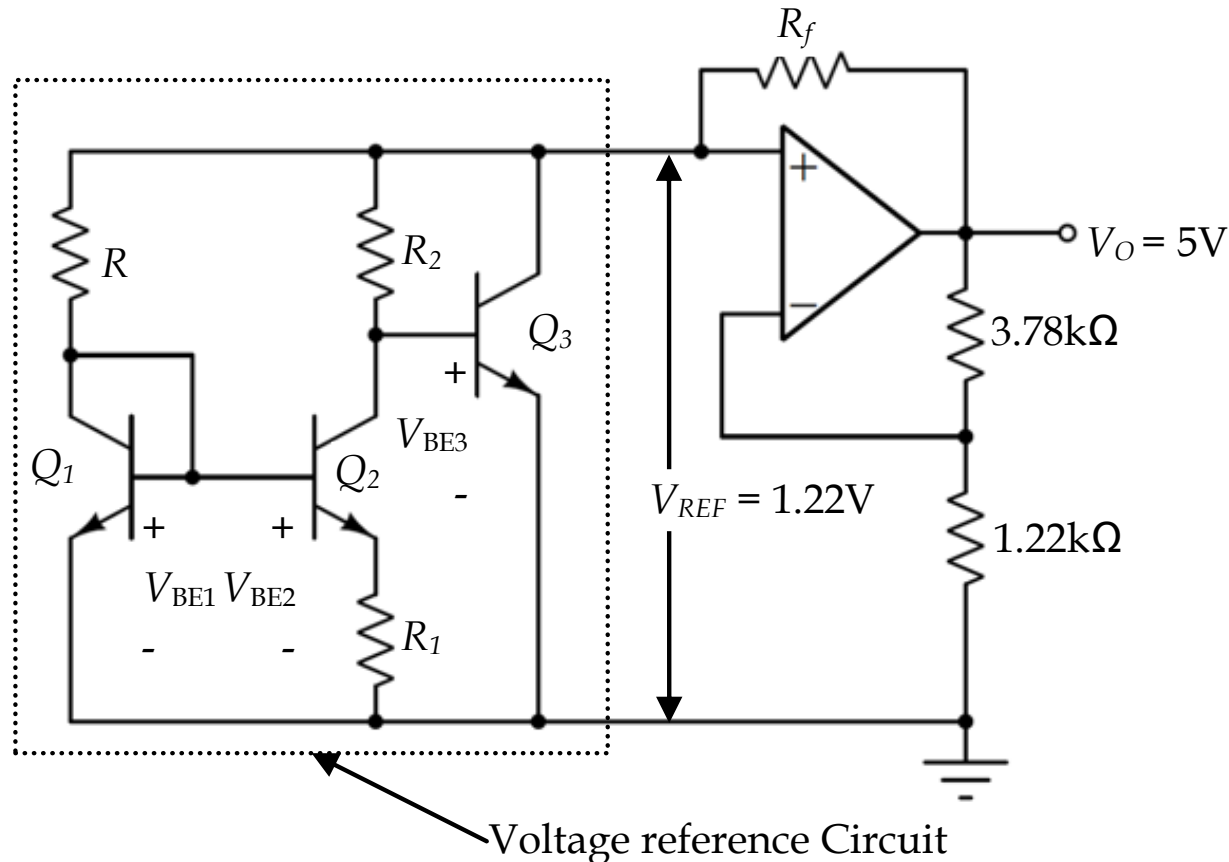


- ✓ The circuit diagram implementation of regulated DC power supply system is:



BANDGAP REFERENCE VOLTAGE SOURCE

- ✓ Bandgap reference voltage source is a circuit which functions like a precise zener diode. It generates a reference voltage that has a zero temperature coefficient characteristics. It implies that the DC output voltage will be independent of variations in temperature.
- ✓ The circuit diagram for bandgap reference source for 5V DC output is given below:



From the circuit,

$$I_{C2} = \frac{V_{BE1} - V_{BE2}}{R_1}$$

$$\text{or, } V_{BE1} - V_{BE2} = I_{C2} R_1 = V_T \ln \left(\frac{I_{C1}}{I_{C2}} \right) \quad \text{Derivation}$$

$$\therefore I_{C2} \propto V_T$$

Since, I_{C2} is directly proportional to thermal voltage, it has positive temperature coefficient characteristics. Thus, the voltage drop across resistor R_2 will have positive temperature coefficient characteristics.

The base emitter voltage drop (V_{BE3}) of transistor Q_3 have negative temperature coefficient characteristics. This voltage (V_{BE3}) when added with the voltage drop across resistor R_2 , i.e. $I_{C2} R_2$ the resultant voltage V_{REF} is generated. V_{REF} will have zero temperature coefficient characteristics. This is because the +ve temperature coefficient characteristics of voltage drop across R_2 cancels out the -ve temperature coefficient characteristics of the resultant reference voltage V_{REF} .

$$V_{REF} = V_{BE3} + I_{C2} R_2 = V_{BE3} + \frac{V_{BE1} - V_{BE2}}{R_1} R_2$$

The bandgap reference voltage source is designed to have

$$V_{REF} = 1.22 \text{ V}$$

This source is called bandgap reference voltage source because the magnitude of V_{REF} is equal to the magnitude of bandgap energy level of silicon which is 1.22 eV.

The output voltage is calculated using following formula,

$$V_{REF} = V_0 \left(\frac{R_y}{R_x + R_y} \right)$$
$$\therefore V_0 = V_{REF} \left(1 + \frac{R_x}{R_y} \right) = 1.22 \times \left(1 + \frac{R_x}{R_y} \right)$$

Choosing $R_x = 3.78 \text{ k}\Omega$ and $R_y = 1.22 \text{ k}\Omega$, we get,

$$V_0 = 5 \text{ V}$$

SERIES VOLTAGE REGULATORS

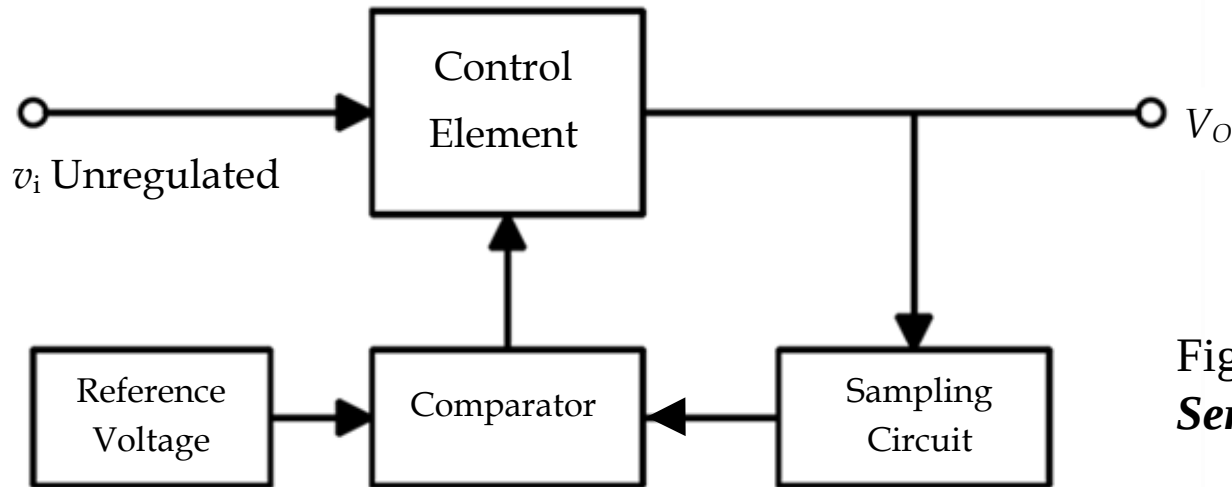


Fig: **Block diagram of Series Voltage Regulator**

- ✓ Figure shows the block diagram of series voltage regulator. The series control element controls the amount of input that gets to the output. The output voltage is sampled by a circuit that provides a feedback voltage to be compared to a reference voltage.
- ✓ If the output voltage increases, the comparator circuit provides control signal to the series control element to decrease the amount of output voltage thereby maintaining the output voltage.
- ✓ If the output voltage decreases, the comparator circuit provides control signal to the series control element to increase the amount of output voltage thereby maintaining the output voltage.

TRANSISTOR SERIES REGULATOR

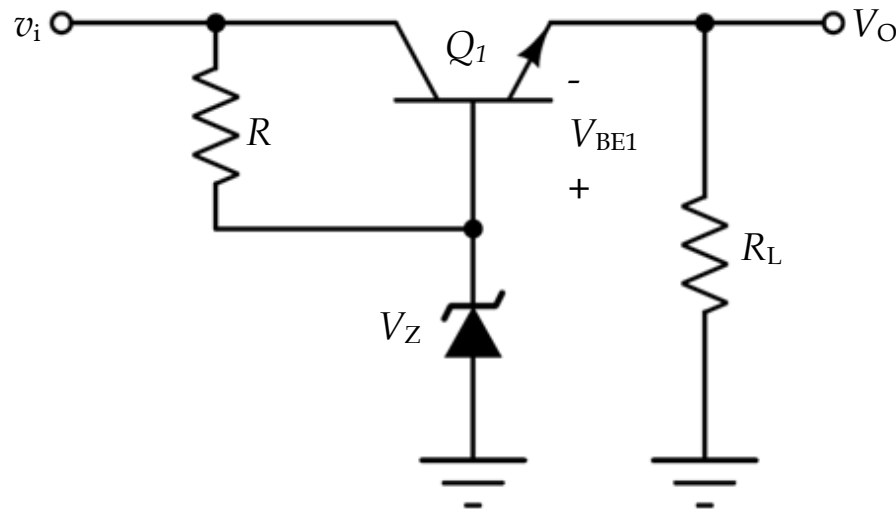


Fig: *Circuit diagram of Transistor Series Regulator*

- ✓ Figure shows the circuit diagram of simple transistor series regulator. In this circuit zener diode provides reference voltage. Where as transistor Q_1 forms the series control element.
- ✓ When output voltage decreases, the base emitter voltage drop of Q_1 is increased, as a result transistor Q_1 conducts more thereby raising the output voltage.
- ✓ When output voltage increases, the base emitter voltage drop of Q_1 is decreased, as a result transistor Q_1 conducts less thereby reducing the output voltage.
- ✓ Hence, the net effect is that the output voltage remains constant or regulated.

IMPROVED TRANSISTOR SERIES REGULATOR

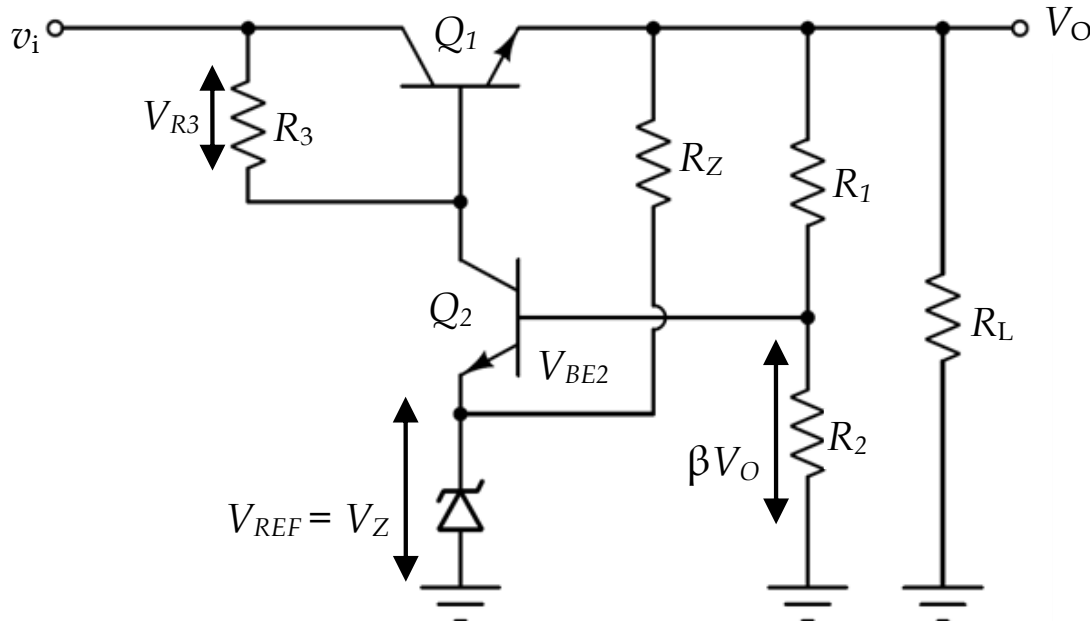


Fig: *Circuit diagram of Improved Transistor Series Regulator*

- ✓ Figure shows the circuit diagram of improved transistor series regulator. In this circuit resistors R_1 and R_2 act as a sampling network with zener diode providing a reference voltage. The transistor Q_2 controls the base current to the transistor Q_1 to vary the current passed by transistor Q_1 to maintain the output voltage constant.
- ✓ If the output voltage tries to increase, the voltage sampled by R_1 and R_2 increases. The increased voltage βV_O causes the base emitter voltage of transistor Q_2 to go up. Thus transistor Q_2 conducts more reducing the base current of transistor Q_1 which in turn passes less current to the load thereby maintaining the output voltage.

If the output voltage tries to decrease, the reverse happens. The voltage sampled by R_1 and R_2 decreases. Thus βV_0 is decreased which decreases the base emitter voltage of transistor Q_2 . As a result transistor Q_2 conducts less increasing the base current to the transistor Q_1 which in turn passes more current to the load thereby maintaining the output voltage.

From the circuit diagram,

$$V_{BE2} + V_Z = \beta V_0 = \frac{R_2}{R_1 + R_2} V_0$$

Re-arranging the above equation, we get,

$$V_0 = \frac{R_1 + R_2}{R_2} (V_{BE2} + V_Z)$$
$$\therefore V_0 = \left(1 + \frac{R_1}{R_2}\right) (V_{BE2} + V_Z)$$

CURRENT LIMITING CIRCUIT

Current limiting circuit is used for the short circuit protection. Figure shows the circuit diagram of current limiting circuit. As the load current I_L increases, the voltage drop across the short circuit sensing resistor R_{SC} is also increased. As a result transistor Q_2 turns on diverting the base current to transistor Q_1 . Thus Q_1 conduct less and load current through Q_1 is reduced preventing any additional current to the load R_L . Hence the action of R_{SC} and transistor Q_2 provides limiting of the maximum load current.

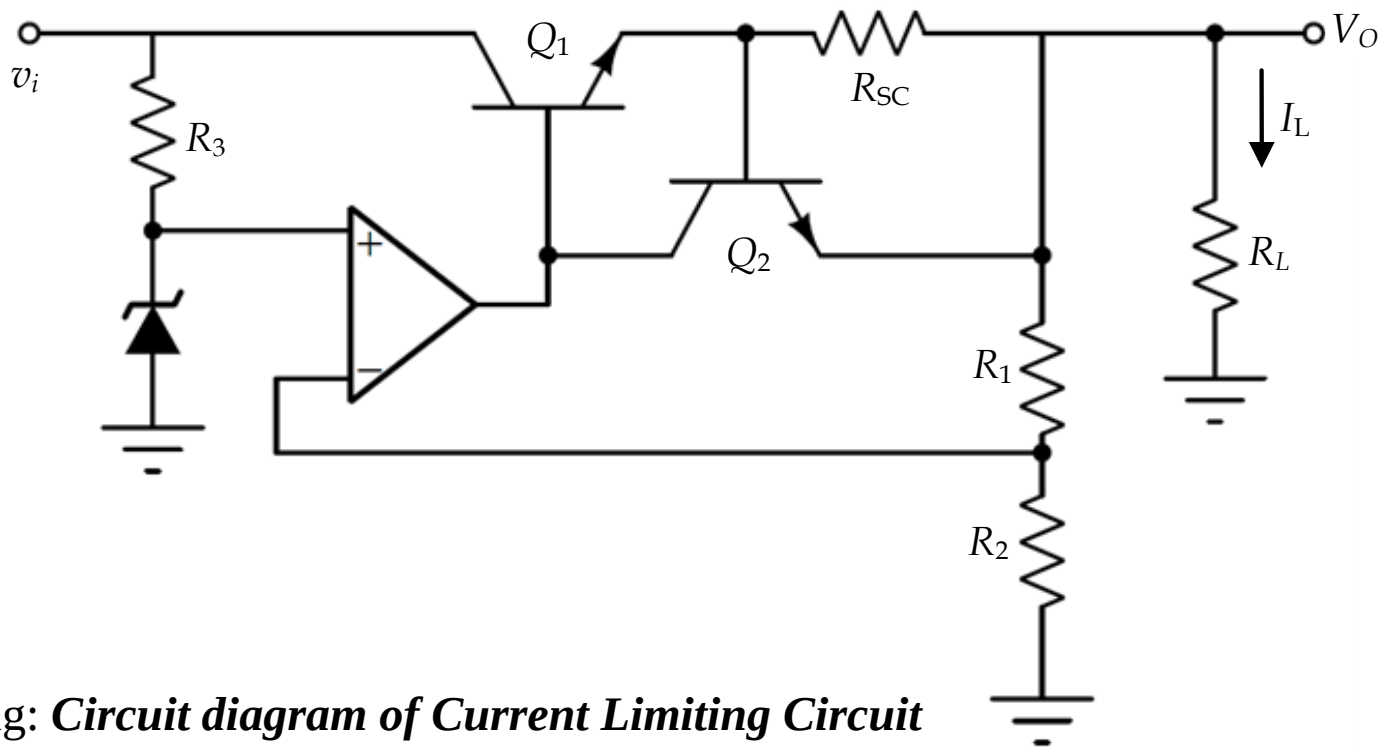


Fig: *Circuit diagram of Current Limiting Circuit*

IC VOLTAGE REGULATORS

The voltage regulator circuits which are available in the form of single IC are known as IC voltage regulators. Generally IC voltage regulators are available as the three terminal IC voltage regulators.

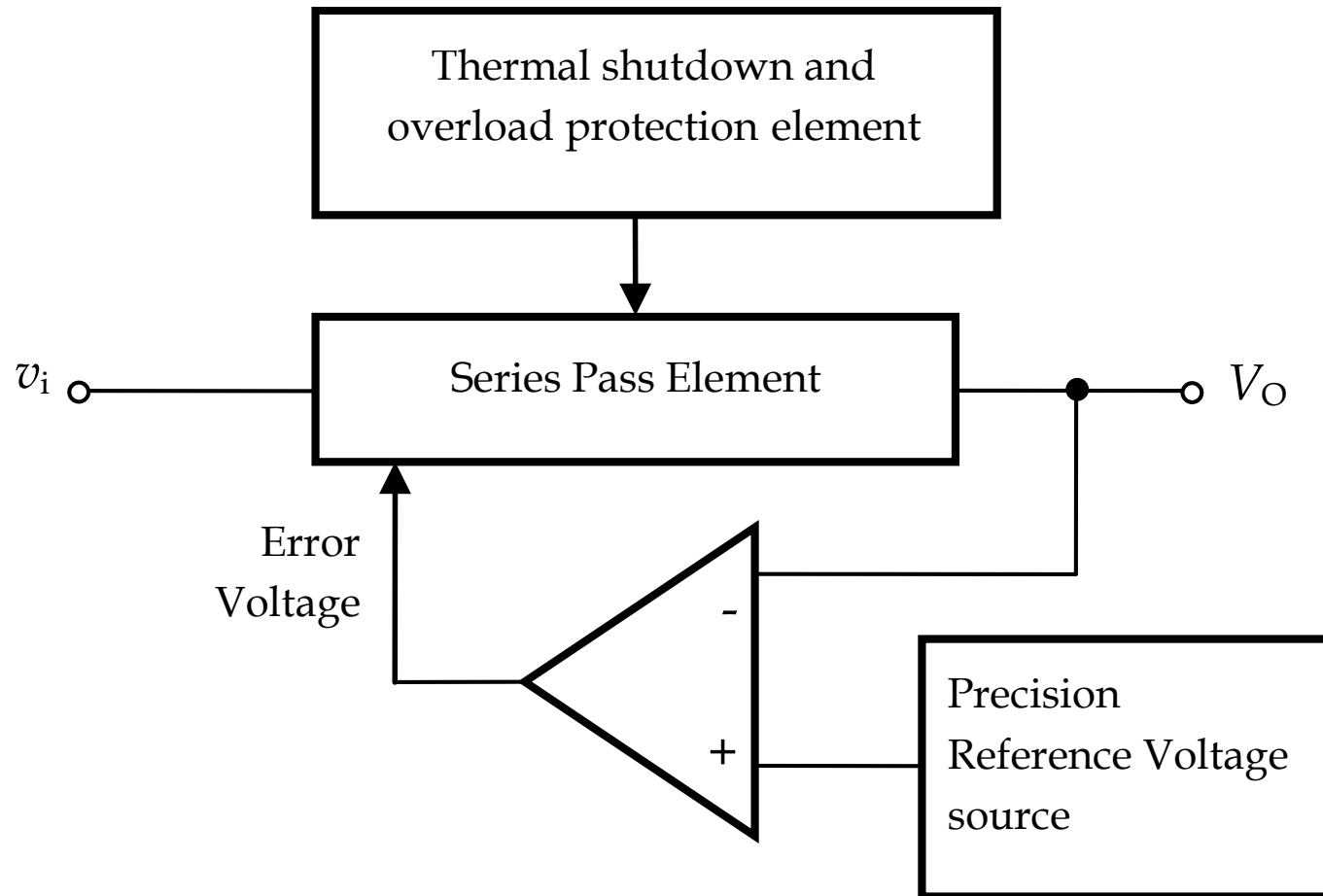


Fig: *Block diagram of IC voltage regulators*

FIXED POSITIVE IC VOLTAGE REGULATORS

The 78XX series regulators provide fixed regulated positive voltages from the 5V to 24V. Let us consider an IC 7805 from 78XX series. This IC provides regulated output of 5V. IC 7805 can be connected as follows so as to get a regulated positive output voltage of 5V.

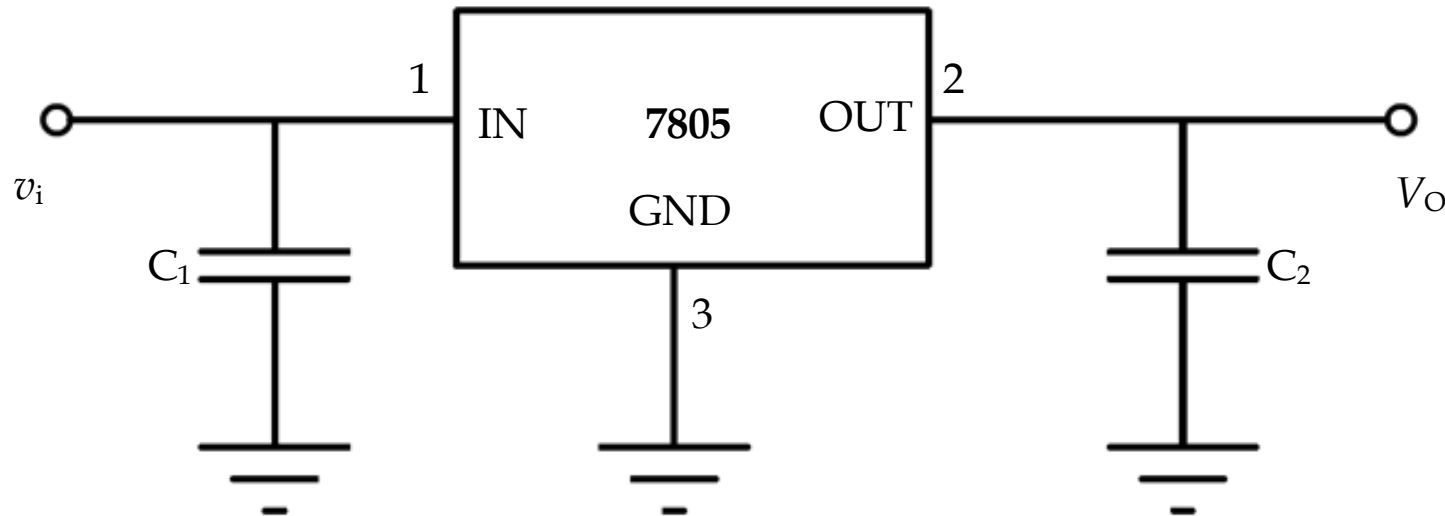


Fig: *Use of 7805 IC to get +5 V DC output*

Table : Positive voltage regulators in the 78XX series

IC	Output Voltage(V_o)	Minimum input(v_i)
7805	+5	7.3
7806	+6	8.3
7808	+8	10.5
7810	+10	12.5
7812	+12	14.6
7815	+15	17.7
7818	+18	21.0
7824	+24	27.1

FIXED NEGATIVE IC VOLTAGE REGULATORS

The 79XX series regulators provide fixed regulated negative voltages from the -5V to -24V. Let us consider an IC 7905 from 79XX series. This IC provides regulated output of - 5V. IC 7905 can be connected as follows so as to get a regulated negative output voltage of -5V.

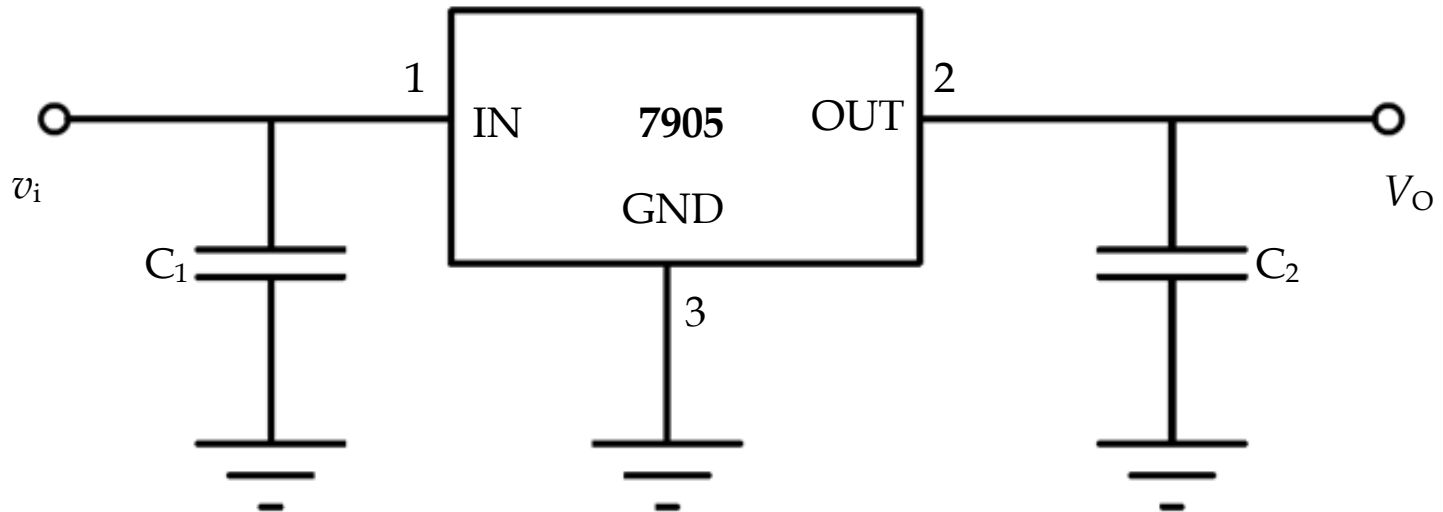


Fig: *Use of 7905 IC to get -5 V DC output*

Table : Negative voltage regulators in the 79XX series

IC	Output Voltage(V_o)	Minimum input(v_i)
7905	-5	-7.3
7906	-6	-8.3
7908	-8	-10.5
7910	-10	-12.5
7912	-12	-14.6
7915	-15	-17.7
7918	-18	-21.0
7924	-24	-27.1

ADJUSTABLE IC VOLTAGE REGULATORS

The voltage regulators that allow the user to set the output voltage to a desired regulated value are known as adjustable voltage regulator. LM 317, for example can be operated with the output voltage regulated at any setting over the range of voltage from 1.2 V to 37 V. Figure below shows how the regulated output voltage of LM 317 can be set.

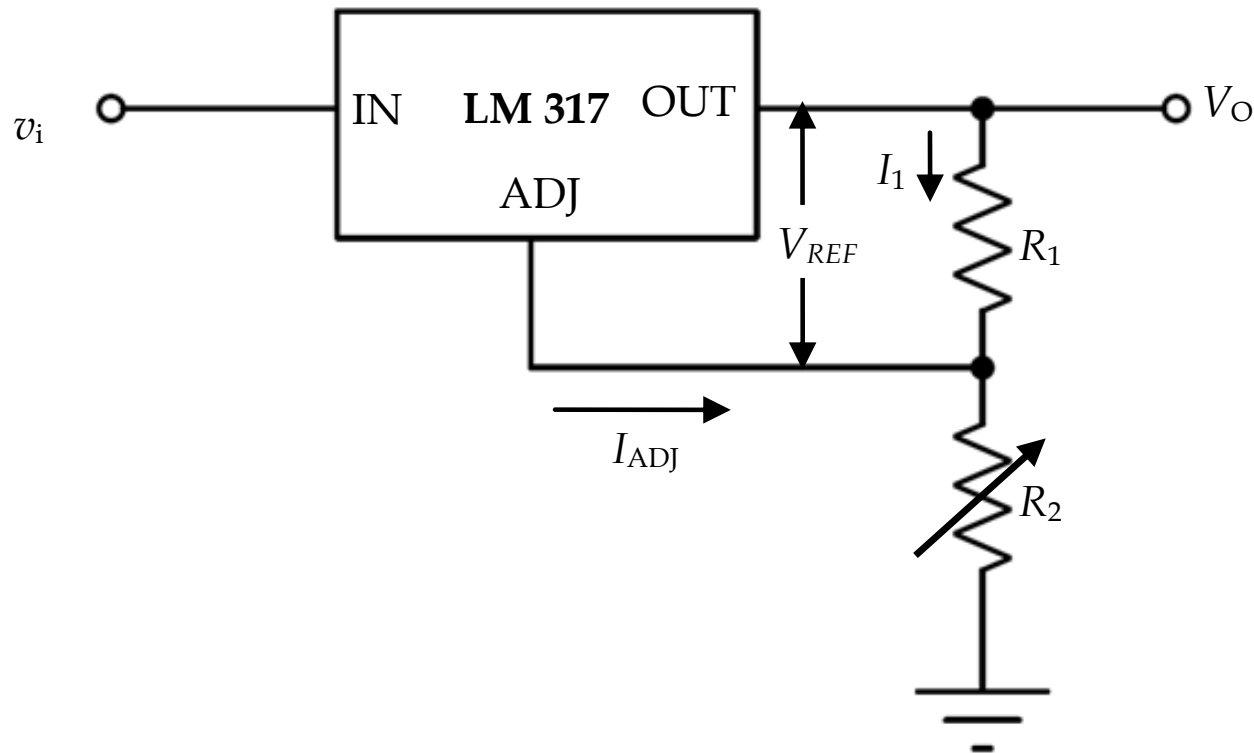


Fig: *Adjustable IC voltage regulator using LM 317*

In the above circuit resistors R_1 and R_2 sets the output to any desired voltage over the adjustment range (1.2V to 37 V). The output voltage desired can be calculated using

$$V_0 = V_{REF} \left(1 + \frac{R_2}{R_1} \right) + I_{ADJ} \times R_2$$

The typical values of V_{REF} and I_{ADJ} are

$V_{REF} = 1.25 \text{ V}$ and $I_{ADJ} = 100 \mu\text{A}$

Since, I_{ADJ} is very small, the voltage drop across resistor R_2 can be ignored,

$$\therefore V_0 = V_{REF} \left(1 + \frac{R_2}{R_1} \right)$$

PERFORMANCE PARAMETERS OF VOLTAGE REGULATORS

Voltage Stability Factor(S_V):

It is defined as the ratio of change in the output voltage to the change in input voltage for a constant load current and temperature. It is denoted as S_V .

Mathematically,

$$S_V = \frac{\Delta V_O}{\Delta v_i} \text{ at const. } I_L \text{ and } T$$

Temperature Stability Factor(S_T):

It is defined as the ratio of change in output voltage to the change in temperature at constant load current and input voltage. It is denoted as S_T .

Mathematically,

$$S_T = \frac{\Delta V_O}{\Delta T} \text{ at const. } I_L \text{ and } v_i$$

Output Resistance(R_O):

It is defined as the ratio of the change in the output voltage to the change in the load current for constant input voltage and temperature. It is denoted as R_O .

Mathematically,

$$R_O = \frac{\Delta V_O}{\Delta I_L} \text{ at const. } v_i \text{ and } T$$

Practically, all the parameters S_V , S_T , R_O should be as low as possible. Thus, the total change in the output voltage is given as:

$$\Delta V_O = S_V \Delta v_i + S_T \Delta T + R_O \Delta I_L$$

DERIVATION OF S_V FOR SERIES VOLTAGE REGULATOR

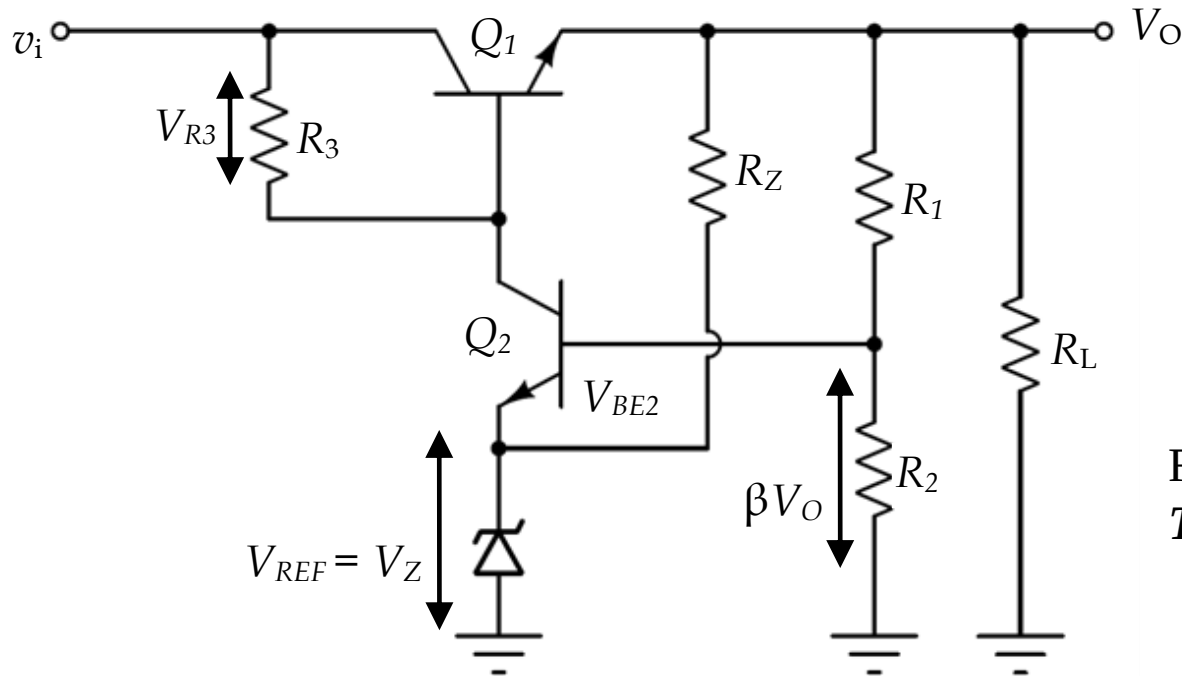


Fig: *Circuit diagram of Improved Transistor Series Regulator*

From the circuit, application of KVL results,

$$v_i = V_{R3} + V_{BE1} + V_O$$

Since the base emitter voltage drop of transistor Q_1 is a constant quantity and output is desired to be constant, we may assume the change in input voltage is approximately equal to the change in voltage across the resistor R_3 when current through it varies. $\Delta v_i = \Delta V_{R3}$

The gain A of the transistor Q_2 is expressed as:

$$A = \frac{\text{Total resistance in the collector of } Q_2}{\text{Total resistance in the emitter of } Q_2}$$

$$\therefore A = \frac{R_3}{r_{e2} + r_Z}$$

Since the feedback voltage βV_O is input to the base of Q_2 and V_{R3} is the output of Q_2 . Therefore we may write,

$$V_{R3} = A \times \beta V_O$$

Let ΔV_O be the small change in the output voltage and ΔV_{R3} be the small change in the voltage across the resistor R_3 .

Then,

$$\Delta V_{R3} = A \times \beta \Delta V_O$$

$$\text{or, } \Delta V_{R3} = \frac{R_3}{r_{e2} + r_Z} \times \beta \times \Delta V_O$$

$$\text{or, } \frac{\Delta V_O}{\Delta V_{R3}} = \frac{r_{e2} + r_Z}{R_3} \times \frac{1}{\beta}$$

$$\text{or, } \frac{\Delta V_O}{\Delta v_i} = \frac{r_{e2} + r_Z}{R_3} \times \frac{1}{\left(\frac{R_2}{R_1 + R_2} \right)}$$

$$\therefore S_V = \left(\frac{r_{e2} + r_Z}{R_3} \right) \left(1 + \frac{R_1}{R_2} \right)$$

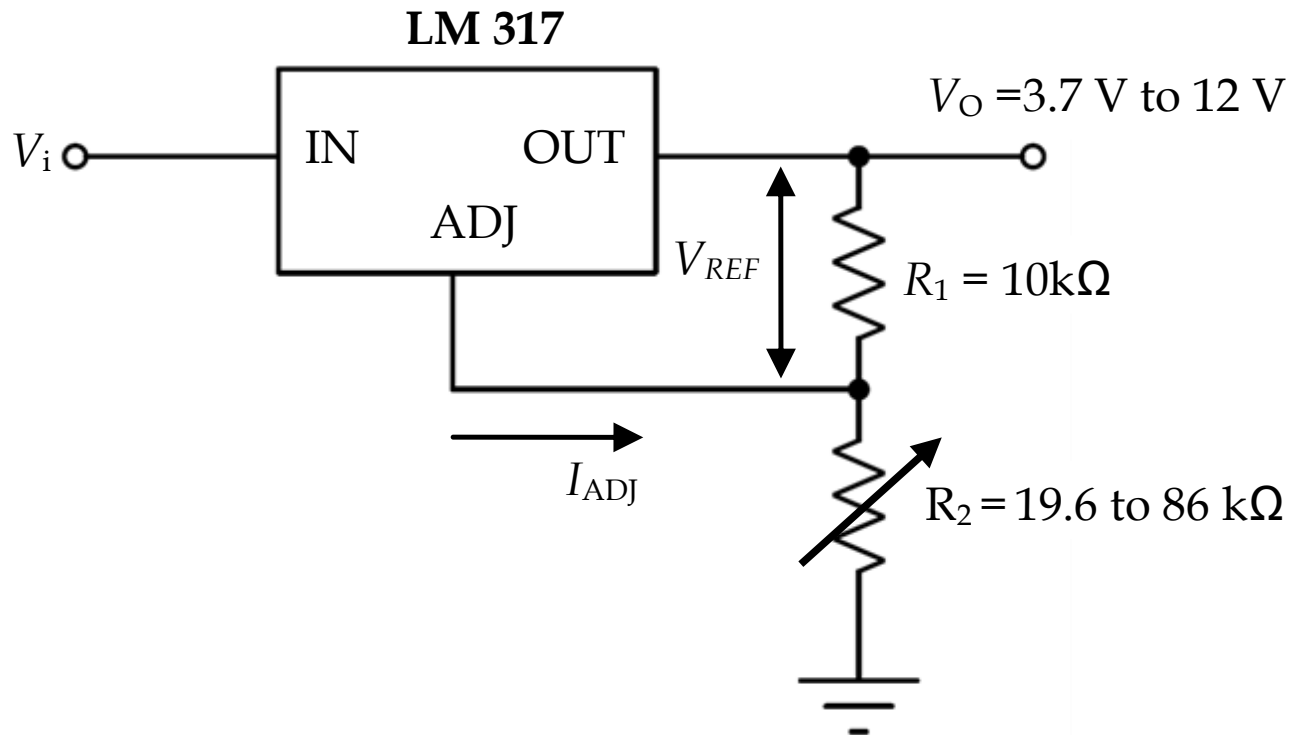
NUMERICALS

Q₁. Design a DCV regulator for 3.7 V to 12 V output using LM317.

2073 Shrawan

Solution:

Here we have to design a voltage regulator with adjustable output requirement. Therefore we shall use LM317 adjustable IC voltage regulator.



We know,

$$V_0 = V_{REF} \left(1 + \frac{R_2}{R_1} \right) = 1.25 \left(1 + \frac{R_2}{R_1} \right)$$

Re-arranging the above equation, we get,

$$R_2 = R_1 \left(\frac{V_0}{1.25} - 1 \right)$$

For $V_0 = 3.7$ V,

$$R_2 = R_1 \left(\frac{3.7}{1.25} - 1 \right) = 1.96R_1$$

$$\therefore R_2 = 1.96R_1$$

For $V_0 = 12$ V,

$$R_2 = R_1 \left(\frac{12}{1.25} - 1 \right) = 8.6R_1$$

$$\therefore R_2 = 8.6R_1$$

Taking $R_1 = 10 \text{ k}\Omega$,

For $V_O = 3.7 \text{ V}$,

$$R_2 = 1.96 R_1 = 1.96 \times 10 = 19.6 \text{ k}\Omega$$

For $V_O = 12 \text{ V}$,

$$R_2 = 8.6 R_1 = 8.6 \times 10 = 86 \text{ k}\Omega$$

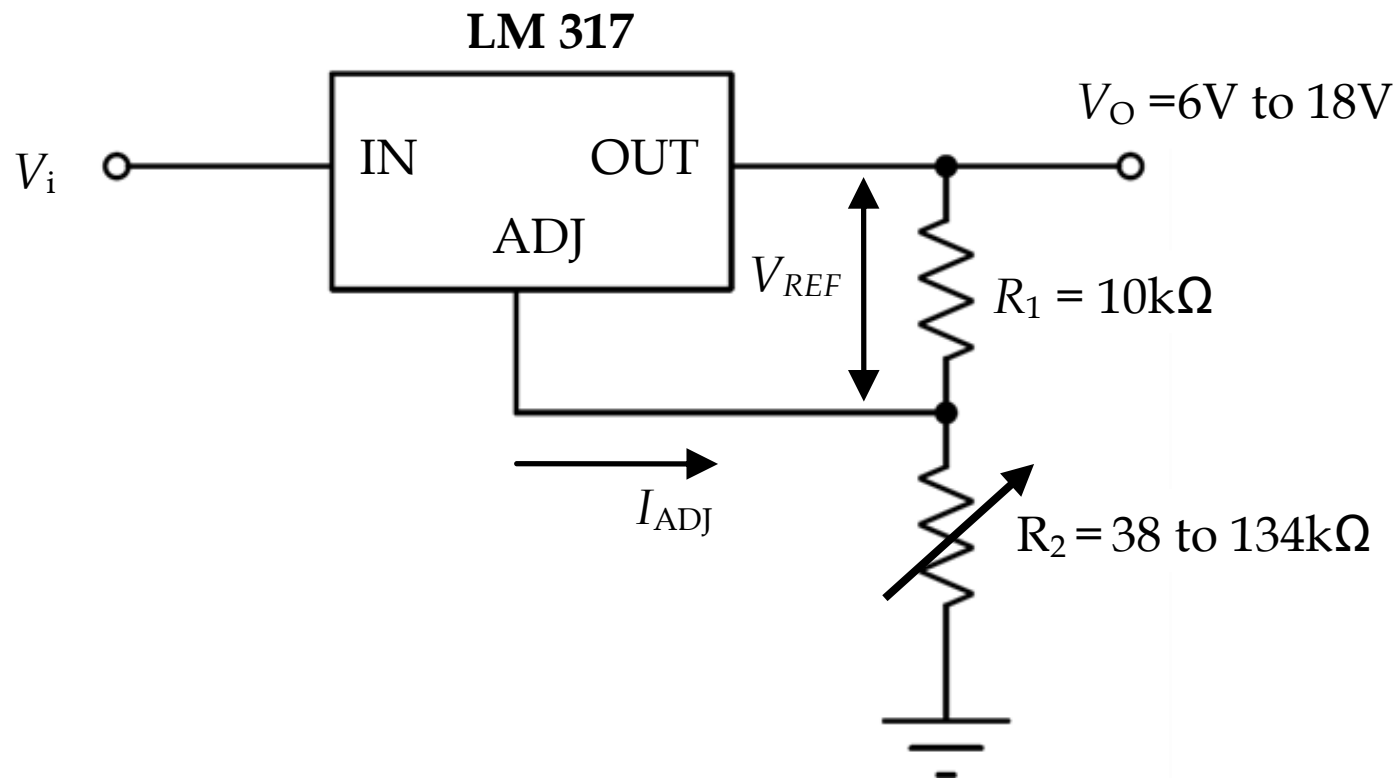
Thus, if we take $R_1 = 10 \text{ k}\Omega$, then the output voltage can be obtained in the range $V_O = 3.7 \text{ V}$ to 12 V by taking a variable resistor R_2 having resistance in the range $R_2 = 19.6 \text{ k}\Omega$ to $86 \text{ k}\Omega$.

Q₂. Design a dc voltage regulator for $V_O = 6V$ to $18V$.

2071 Chaitra

Solution:

Use similar process as in Q₁ to obtain following answer:



Q₃. Design a (10-25) V variable DC series voltage regulator using LM317 IC.
2070 Chaitra

Solution:

Use similar process as in Q₁ to obtain following answer:

